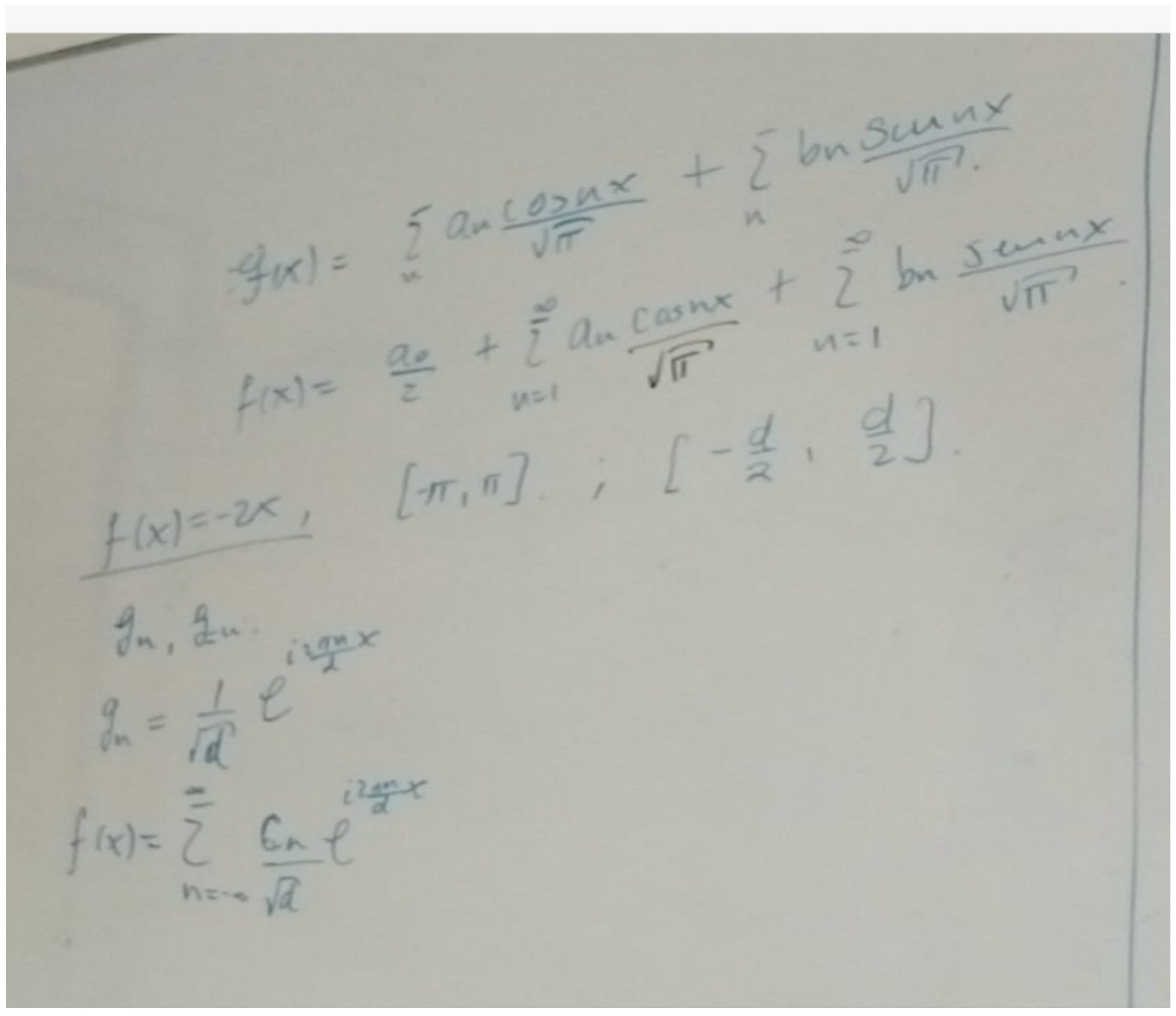
Funcions especials — clase # 14 — 02/09/2025

$$f(x) = \sum_n a_n \frac{\cos nx}{\sqrt{\pi}} + \sum_n b_n \frac{\sin nx}{\sqrt{\pi}}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \frac{\cos nx}{\sqrt{\pi}} + \sum_{n=1}^{\infty} b_n \frac{\sin nx}{\sqrt{\pi}}$$

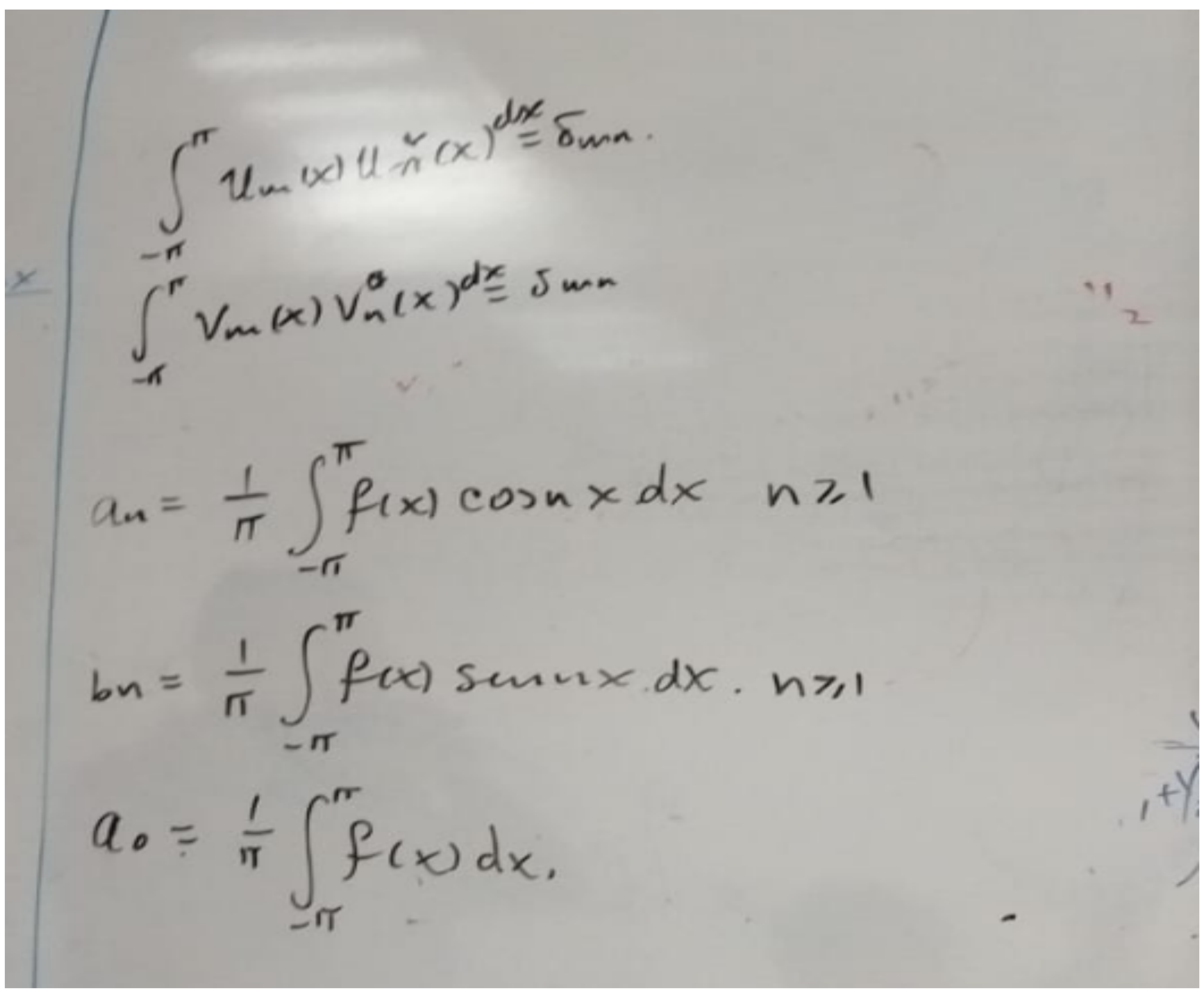
$$\boxed{f(x) = -2x}, \quad [-\pi, \pi] \; ; \; \left[-\frac{d}{2}, \frac{d}{2}\right]$$



g_n, g_{-n}

$$g_n = \frac{1}{\sqrt{d}} e^{i\frac{2\pi n x}{d}}$$

$$f(x) = \sum_{n=-\infty}^{\infty} \frac{C_n}{\sqrt{d}} e^{i\frac{2\pi n x}{d}}$$



Tenemos...

$$f(x) = -2x \; ; \qquad a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} (-2x)\,dx = 0$$

$$\boxed{a_n = 0}$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} (-2x)\,\text{sen}\,nx\,dx$$

Tenemos

$$b_n = -\frac{2}{\pi}\int_{-\pi}^{\pi} x\,\text{sen}\,nx\,dx \; - \; \int x\,\frac{d(\cos nx)}{n}$$

$\vdots$

$=$

$$a_n = 0.$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} (-2x)\,\text{sen}\,nx\,dx.$$

$$b_n = -\frac{2}{\pi}\int_{-\pi}^{\pi} x\,\text{sen}\,nx\,dx. \; - \int x\,\frac{d(\cos nx)}{n}$$

$$= \frac{2}{n\pi}\left\{ x\cos nx \Big|_{-\pi}^{\pi} - \int_{-\pi}^{\pi} \underline{\cos nx\,dx} \right\}.$$

$$\frac{\text{sen}\,nx}{n}\Big|_{-\pi}^{\pi}$$

Tenemos ...

$$\frac{2}{n\pi}\left[\pi \cos n\pi + \pi \cos(n\pi)\right]$$

$\vdots$

$$\frac{4}{n\pi}\left[\cos n\pi\right] \begin{cases} \bullet\ \dfrac{4}{n\pi} & \text{si } n \text{ es par} \\ \bullet\ -\dfrac{4}{n\pi} & \text{si } n \text{ es impar} \end{cases}$$

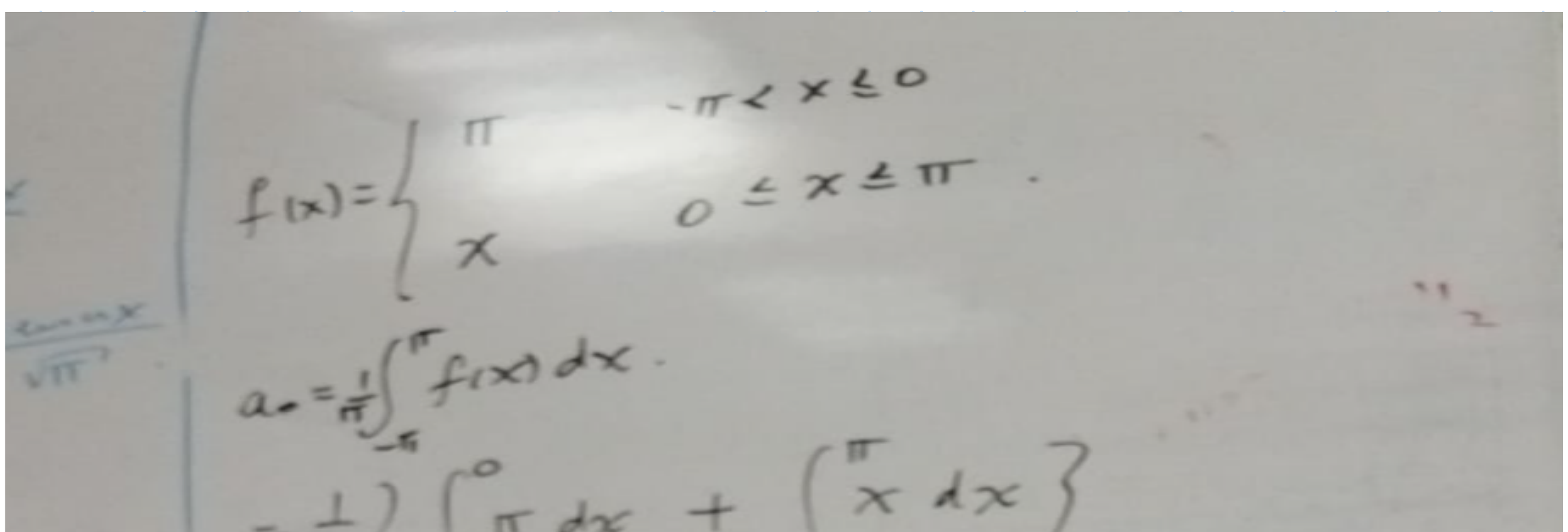
Tenemos

$$\Rightarrow \boxed{-2x = \sum_{n=1}^{\infty} \frac{4}{n\pi}(-1)^n \operatorname{Sen} nx}$$

Siguiente caso:

$$f(x) = \begin{cases} \pi & -\pi < x \leq 0 \\ x & 0 \leq x \leq \pi \end{cases}$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx$$



$$f(x) = \begin{cases} \pi & -\pi < x \leq 0 \\ x & 0 \leq x \leq \pi \end{cases}$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx$$

$$= \frac{1}{\pi}\left[\pi^2 + \frac{\pi^2}{2}\right] = \frac{1}{\pi}\frac{3\pi^2}{2} = \frac{3\pi}{2}.$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx.$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} \pi\cos nx\,dx + \int_{0}^{\pi} x\cos nx\,dx\right\}$$

$$= \frac{1}{\pi}\left\{\cancel{\pi\,\frac{\operatorname{sen} nx}{n}\Big|_{-\pi}^{0}}^{\,0} + \int_0^{\pi} x\,\frac{d(\operatorname{sen} nx)}{n}\right\}.$$

$$= \frac{1}{\pi}\left\{\cancel{\frac{x\operatorname{sen} nx}{n}\Big|_0^{\pi}}^{\,0} - \int_0^{\pi}\frac{\operatorname{sen} nx\,dx}{n}\right\}.$$

$$\frac{1}{\pi}\left\{\frac{\cos nx}{n^2}\right\}\Big|_0^{\pi} = \frac{1}{\pi n^2}\cos nx\Big|_0^{\pi}\begin{cases}0 & n \text{ par}\\ -2 & n \text{ impar}\end{cases}$$

-1

$\frac{2}{n\pi}\left[\pi c\ldots\right.$

$\frac{4}{-n\pi}\left[c\ldots\right.$

$$a_n = \begin{cases} 0\ ; & n \text{ par} \\ -\frac{2}{\pi n^2}\ ; & n \text{ impar}\end{cases}$$

Ahora para b_n:

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\operatorname{sen} nx\,dx$$

$$b_n = \frac{1}{\pi}\left\{\int_{-\pi}^{0}\pi\operatorname{sen} nx\,dx + \int_0^{\pi} x\operatorname{sen} nx\,dx\right\}$$

$$-\pi \frac{\cos nx}{n}\Big|_{-\pi}^{0} - \int_0^\pi \frac{x\, d(\cos nx)}{n}$$

$$-\frac{1}{n}\left\{ x\cos nx \Big|_0^\pi - \int \cos nx\, dx \right\}$$

$$-\frac{1}{n}\left\{ x\cos nx \Big|_0^\pi - \frac{\operatorname{sen} nx}{n}\Big|_0^\pi \right\}$$

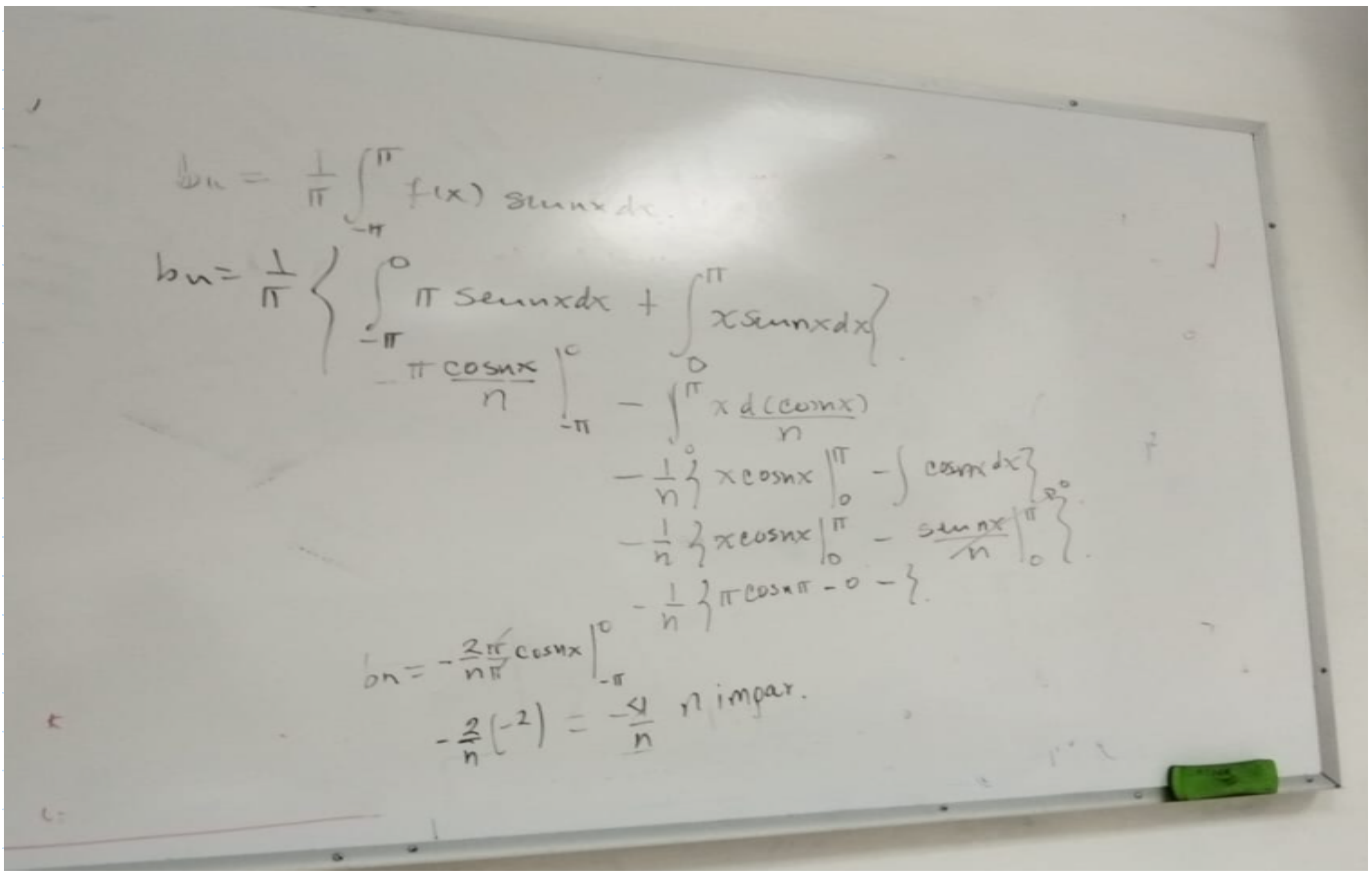
$$-\frac{1}{n}\left\{ \pi\cos n\pi - 0 \right\}$$

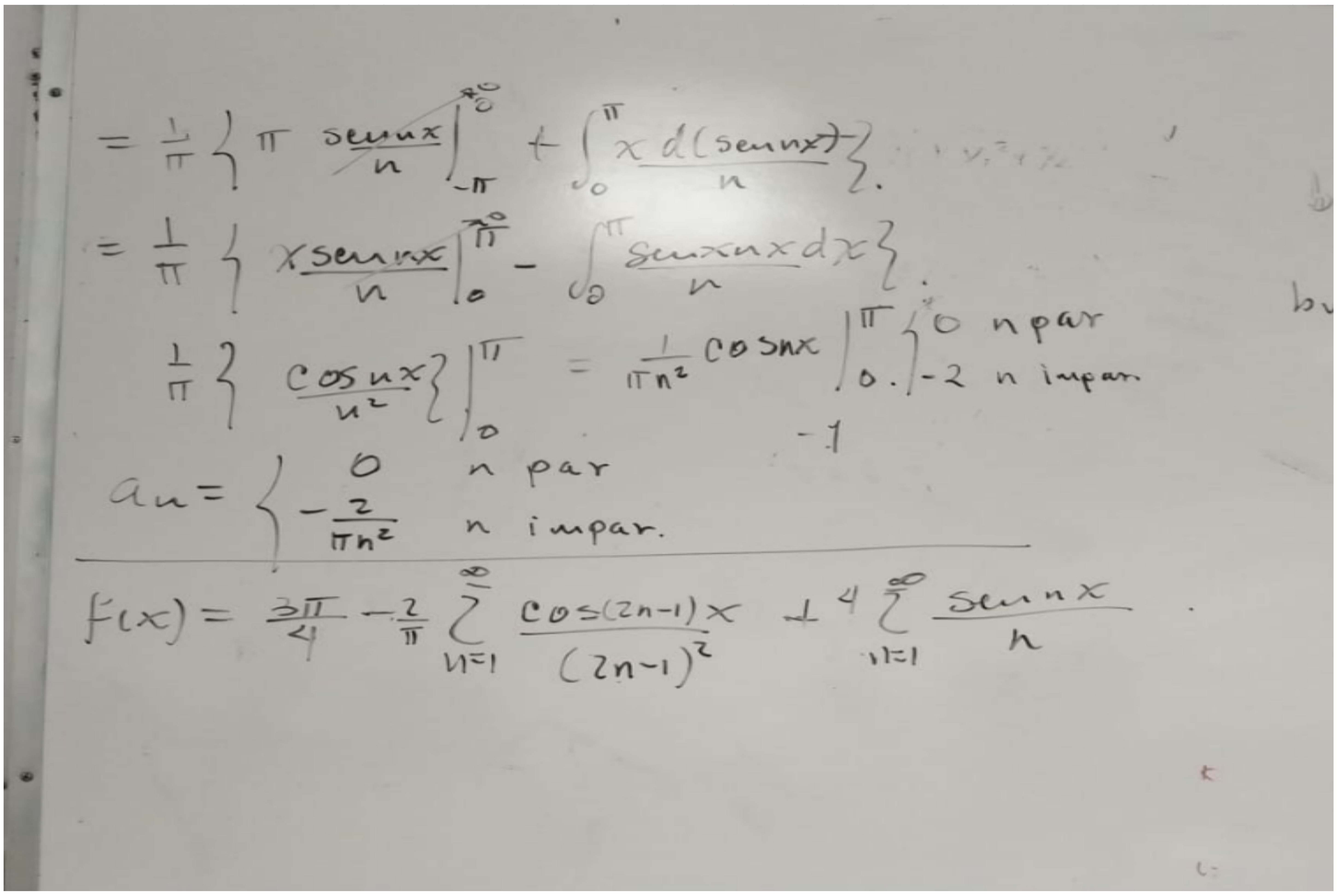
Tenemos...

$$b_n = -\frac{2}{n}\frac{\pi}{\pi}\cos nx\Big|_{-\pi}^{0}$$

$$\vdots$$

$$= -\frac{2}{n}(-2) = -\frac{4}{n}\ ; \quad n \text{ impar}$$





$$= \frac{1}{\pi}\left\{\pi \frac{\sin nx}{n}\Big|_{-\pi}^{0} + \int_0^\pi x\, d\left(\frac{\sin nx}{n}\right)\right\}.$$

$$= \frac{1}{\pi}\left\{\frac{x \sin nx}{n}\Big|_0^\pi - \int_0^\pi \frac{\sin nx\, dx}{n}\right\}.$$

$$\frac{1}{\pi}\left\{\frac{\cos nx}{n^2}\right\}\Big|_0^\pi = \frac{1}{\pi n^2}\cos nx\Big|_0^\pi \begin{cases} 0 & n \text{ par} \\ -2 & n \text{ impar} \end{cases}$$

-1

$$a_n = \begin{cases} 0 & n \text{ par} \\ -\frac{2}{\pi n^2} & n \text{ impar.} \end{cases}$$

$$f(x) = \frac{3\pi}{4} - \frac{2}{\pi}\sum_{n=1}^{\infty} \frac{\cos(2n-1)x}{(2n-1)^2} + 4\sum_{n=1}^{\infty}\frac{\sin nx}{n}.$$

Tenemos ahora: Ecuación de Helmholtz en coordenadas cilíndricas

Más general que Ec. de Laplace

$$\phi(x,y,z) = X(x)\,Y(y)\,Z(z)$$

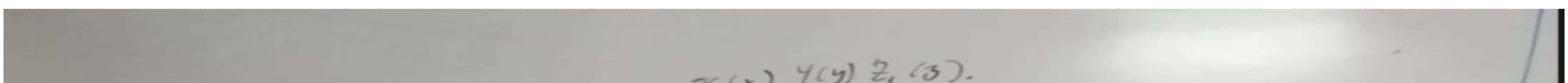
$$\Rightarrow \quad \nabla^2\phi + k^2\phi = 0$$

Tenemos

$$\frac{1}{\rho}\left[\frac{\partial}{\partial\rho}\left(\frac{\rho}{1}\frac{\partial\phi}{\partial\rho}\right) + \frac{\partial}{\partial\varphi}\left(\frac{1}{\rho}\frac{\partial\phi}{\partial\varphi}\right) + \frac{\partial}{\partial z}\left(\frac{\rho}{1}\frac{\partial\phi}{\partial z}\right)\right] + k^2\phi = 0$$

$$\Rightarrow \quad \frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial\phi}{\partial\rho}\right) + \frac{1}{\rho^2}\frac{\partial^2\phi}{\partial\varphi^2} + \frac{\partial^2\phi}{\partial z^2} + k^2\phi = 0$$

$$\phi(\rho,\varphi,z) = R(\rho)\,\Phi(\varphi)\,Z(z)$$



$X(x)\,Y(y)\,Z(z).$

$$\phi(x,y,z) = \ldots$$

$$\nabla^2\phi + k^2\phi = 0.$$

$h_1=1,\ h_2=\rho,\ h_3=1.$

$$\frac{1}{\rho}\left[\frac{\partial}{\partial\rho}\left(\frac{\rho}{1}\frac{\partial\phi}{\partial\rho}\right)+\frac{\partial}{\partial\varphi}\left(\frac{1}{\rho}\frac{\partial\phi}{\partial\varphi}\right)+\frac{\partial}{\partial z}\left(\frac{\rho}{1}\frac{\partial\phi}{\partial z}\right)\right]+k^2\phi=0.$$

$$\frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial\phi}{\partial\rho}\right)+\frac{1}{\rho^2}\frac{\partial^2\phi}{\partial\varphi^2}+\frac{\partial^2\phi}{\partial z^2}+k^2\phi=0.$$

$$\phi(\rho,\varphi,z)=R(\rho)\,\Phi(\varphi)\,Z(z)$$

$$\hookrightarrow \frac{\Phi Z}{\rho}\frac{d}{d\rho}\left(\rho\frac{dR}{d\rho}\right)+\frac{RZ}{\rho^2}\frac{d^2\Phi}{d\varphi^2}+R\Phi\frac{d^2Z}{dz^2}+k^2R\Phi Z=0.$$

$$\frac{1}{\rho R}\frac{d}{d\rho}\left(\rho\frac{dR}{d\rho}\right)+\frac{1}{\rho^2\Phi}\frac{d^2\Phi}{d\varphi^2}+\underbrace{\frac{1}{Z}\frac{d^2Z}{dz^2}}_{\ell^2}+k^2=0.$$

$$\nabla^2\phi + k^2\phi = 0.$$

$h_1=1,\ h_2=\rho,\ h_3=1.$

$$\frac{1}{\rho}\left[\frac{\partial}{\partial\rho}\left(\frac{\rho}{1}\frac{\partial\phi}{\partial\rho}\right)+\frac{\partial}{\partial\varphi}\left(\frac{1}{\rho}\frac{\partial\phi}{\partial\varphi}\right)+\frac{\partial}{\partial z}\left(\frac{\rho}{1}\frac{\partial\phi}{\partial z}\right)\right]+k^2\phi=0.$$

$$\frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial\phi}{\partial\rho}\right)+\frac{1}{\rho^2}\frac{\partial^2\phi}{\partial\varphi^2}+\frac{\partial^2\phi}{\partial z^2}+k^2\phi=0.$$

$$\phi(\rho,\varphi,z)=R(\rho)\,\Phi(\varphi)\,Z(z)$$

$$\hookrightarrow \frac{\Phi Z}{\rho}\frac{d}{d\rho}\left(\rho\frac{dR}{d\rho}\right)+\frac{RZ}{\rho^2}\frac{d^2\Phi}{d\varphi^2}+R\Phi\frac{d^2Z}{dz^2}+k^2R\Phi Z=0.$$

$$\frac{1}{\rho R}\frac{d}{d\rho}\left(\rho\frac{dR}{d\rho}\right)+\frac{1}{\rho^2\Phi}\frac{d^2\Phi}{d\varphi^2}+\underbrace{\frac{1}{Z}\frac{d^2Z}{dz^2}}_{\ell^2}+k^2=0.$$

$$\frac{\rho}{R}\frac{d}{d\rho}\left(\rho\frac{dR}{d\rho}\right)$$

$$\frac{1}{\Phi}\frac{d^2\Phi}{d\varphi^2}=-$$

$$\rho\frac{d}{d\rho}\left(\rho\frac{dR}{d\rho}\right.$$

↳ Bes

$$\frac{1}{Z}\frac{d^2Z}{dz^2}=\ell^2.$$

$$\frac{1}{\rho R}\frac{d}{d\rho}\left(\rho\frac{dR}{d\rho}\right)+\frac{1}{\rho^2\Phi}\frac{d^2\Phi}{d\varphi^2}+\underbrace{\ell^2+k^2}_{n^2}=0.$$

$$\frac{\rho}{R}\frac{d}{d\rho}\left(\rho\frac{dR}{d\rho}\right)+\underbrace{\frac{1}{\Phi}\frac{d^2\Phi}{d\varphi^2}}_{-m^2}+n^2\rho^2=0.$$

$$\frac{1}{\Phi}\frac{d^2\Phi}{d\varphi^2}=-m^2.$$

$$\rho\frac{d}{d\rho}\left(\rho\frac{dR}{d\rho}\right)+(n^2\rho^2-m^2)R=0.$$

↳ Bessel.

Siguiente caso...

/nu

$$\kappa^2 = \ell^2 = 0 \; ; \quad m = \nu$$

Tenemos

$$\frac{1}{\phi}\frac{d^2\phi}{d\varphi^2} = -\nu^2$$

$$\Rightarrow \quad \rho \frac{d}{d\rho}\left(\rho \frac{dR}{d\rho}\right) - \nu^2 R = 0$$

si $\nu = 0$

$$\Phi(\varphi) = A_0 + B_0 \Phi$$

$$R(\rho) = C_0 + D_0 \ln \rho$$

Ahora si $\nu \neq 0$

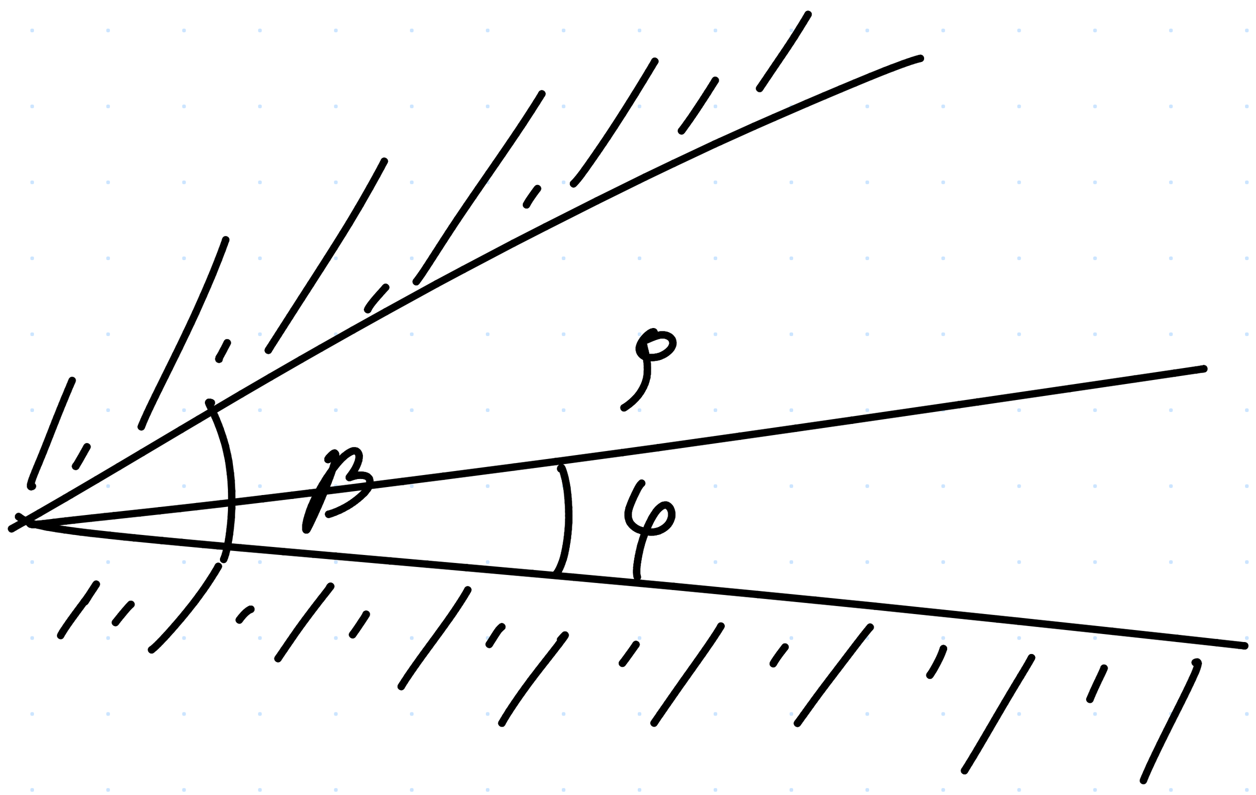
$$\Phi(\varphi) = A_\nu \cos \nu\varphi + B_\nu \operatorname{sen} \nu\varphi$$

$$R(\rho) = C_\nu \rho^\nu + D_\nu \rho^{-\nu}$$

Tenemos

$$\boxed{\phi(\rho,\varphi) = A_0 + b_0 \ln \rho + \sum_\nu \left(a_\nu \rho^\nu + b_\nu \rho^{-\nu}\right)\left(C_\nu \cos\nu\varphi + D_\nu \operatorname{sen}\nu\varphi\right)}$$

Ejemplo



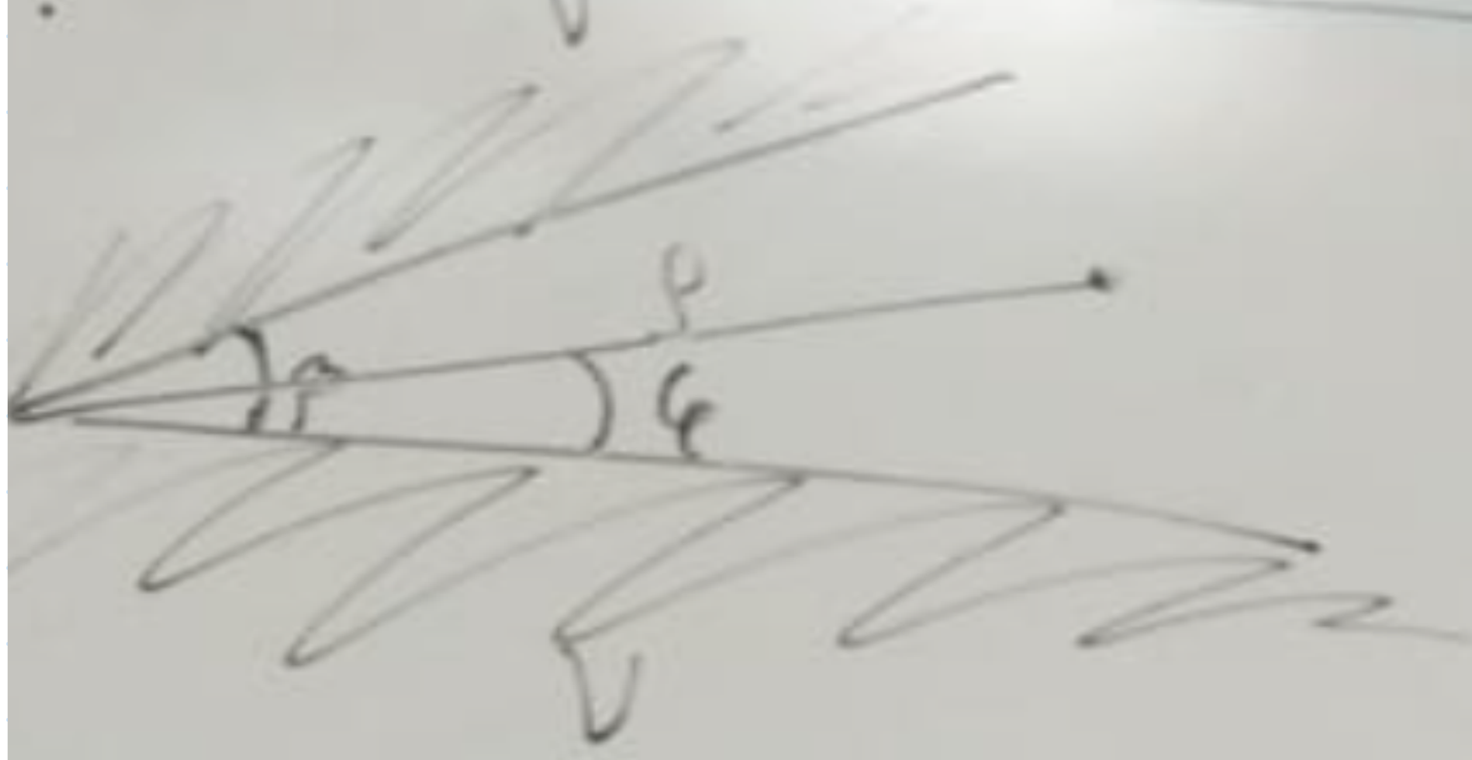
cuando $\varphi = \beta \quad \Phi = V$

$\varphi = 0 ; \quad \Phi = V$

singularidades (evitarlas)

$b_\nu = 0$

$$\Phi(\rho, \varphi) = A_0 + b_0 \ln\rho + \sum_\nu (a_\nu \rho^\nu + b_\nu \rho^{-\nu})(C_\nu \cos\nu\varphi + D_\nu \operatorname{Sen}\nu\varphi)$$



Cuando $\varphi = \beta \quad \Phi = V$.

$\varphi = 0 ; \Phi = V$.

$b_\nu = 0$.

$$\Phi(\rho, \varphi) = A_0 + b_0 \ln\rho + \underbrace{\sum_\nu (C_\nu \cos\nu\varphi + D_\nu \operatorname{Sen}\nu\varphi)\rho^\nu}_{\text{Términos de más alto orden}}$$

Expansión multipolar (Electrodinámica).

Barra infinita (expansión multipolar).

Tenemos

$$\Phi(\rho, \varphi) = A_0 + \cancel{b_0 \ln\rho} + \sum_\nu (C_\nu \cos\nu\varphi + D_\nu \operatorname{Sen}\nu\varphi)\rho^\nu$$

$$\boxed{\Phi(\rho, \varphi) = A_0 + \sum_\nu (C_\nu \cos\nu\varphi + D_\nu \operatorname{Sen}\nu\varphi)\rho^\nu .}$$

Tenemos (condiciones en la frontera)

$$\phi(\rho,0) = V = A_0 + \sum_\nu \left(C_\nu \cos(0) + D_\nu \operatorname{sen}(0)\right).$$

$$A_0 = V. \quad C_\nu = 0.$$

$$\phi(\rho,\varphi) = V + \sum_\nu D_\nu \operatorname{sen} \nu\varphi \, \rho^\nu.$$

$$V = \phi(\rho,\beta) = \cancel{V} + \sum_\nu D_\nu \operatorname{sen} \nu\beta \, \rho^\nu$$

$$\sum_\nu \rho^\nu D_\nu \operatorname{sen} \nu\beta = 0.$$

$$\nu\beta = n\pi.$$

$$\nu = \frac{\pi n}{\beta}$$

$$\boxed{\phi(\rho,\varphi) = V + \sum_\nu D_\nu \operatorname{sen}\left(\frac{\pi n}{\beta}\varphi\right) \rho^{\frac{\pi n}{\beta}}}$$

Expansión en primer orden:

$$\phi(\rho,\varphi) = V + A_0 \operatorname{sen}\left(\frac{\pi\varphi}{\beta}\right) \rho^{\frac{\pi}{\beta}}$$

$$\vec{E} = -\nabla\phi = \frac{\partial\phi}{\partial\rho}\hat{\rho} + \frac{1}{\rho}\frac{\partial\phi}{\partial\varphi}\hat{\varphi}.$$

$$E_\rho = A_0 \frac{\pi}{\beta} \operatorname{sen}\left(\frac{\pi\varphi}{\beta}\right) \rho^{\frac{\pi}{\beta}-1}$$

$$E_\varphi = A_0 \frac{\pi}{\beta} \cos\left(\frac{\pi\varphi}{\beta}\right) \rho^{\frac{\pi}{\beta}-1}$$